MIL-C-453C(OS)
5 November 1976
SUPERSEDING
MIL-C-453B
18 October 1951

MILITARY SPECIFICATION

CORD, LACING, SILK

(FOR ASSEMBLY OF PROPELLING CHARGES)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all the departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for silk lacing cord used in the assembly of propelling charges.

1.2 Classification. Silk lacing cord shall be of the following types and classes, as specified (see 6.1):

Type A--Three cords, each containing not less than 10 single ends
Type B--Braided round without interior axial core
Class a--20-pound strength
Class b--30-pound strength
Class c--90-pound strength
Class d--100-pound strength.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, ATTN: Standardization Division (611), Indian Head, Md. 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4020

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STANDARDS

Military

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Silk lacing cord shall be manufactured from pure, spun or neat, silk.

3.2 Chemical requirements. The silk lacing cord shall meet the requirements of table I.

TABLE I. Chemical requirements.

Property	Requirement
Foreign material	None
Ether extract	0.75%, max
Acidity	
pH of water extract	4.4 to 9.7
Organic or alkalinity	0.1%, max
Ash	1.0%, max

3.3 Physical requirements. Silk lacing cord shall conform to the physical requirements as specified in table II.

TABLE II. Physical requirements.

Property	Class a	Class b	Class c	Class d
Breaking strength, pounds (min.)	20	30	90	100
Yards per pound	1,600 ± 120	850 ± 60	260 ± 20	240 (min)

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3.3.1 Stretch (type B, class d). When tested as specified in 4.3.7, test specimens of type B, class d silk lacing cord shall not stretch more than 8 percent.

3.4 Length. Unless otherwise specified by the procuring activity, silk lacing cord shall be supplied on cones, each containing a minimum of 500 yards.

3.5 Workmanship. The cord shall have a smooth finish, free from knots and slugs to eliminate the possibility of chafing during the lacing operation. When made of three cords, the silk lacing shall have a hard finish.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Size of lot. The size of lot shall be as specified (see 6.1).

4.2.2 Sampling. Ten percent, but nor more than 10 of the boxes of each lot, shall be selected in such a manner as to be representative of the lot. One cone shall be taken from each selected box or bale and at least 20 feet of cord removed before taking samples. Two samples of 15 yards each shall be cut from each cone and marked to indicate the box from which they came. One of each pair of samples shall be held for possible future examination. The composite sample consisting of the remaining pieces shall be marked so as to show the lot number, name of manufacturer, contract or order number, and net weight of the lot. All acceptance tests shall be made on the composite sample representative of the lot.

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4.3 Tests. The tests shall be performed in accordance with the following methods.

4.3.1 Foreign materials. Transfer to each of a number of test tubes 10 millimeters (ml) of a solution prepared by dissolving 175 grams of sodium hydroxide in 825 ml of distilled water. Cut a portion from 1 to 2 inches long from each piece comprising the composite sample and place in the sodium hydroxide solution. Heat the solution almost to boiling. Wool, mohair, and mulberry silk are dissolved at once, while tussah silk dissolves after 1 or 2 minutes of heating or shaking, although a small amount of short hair-like fibers always remain. Cotton will appear unaltered even after 5 minutes of boiling. If this test shows the material to be silk, wool, or mohair, treat threads from each of the pieces in the sample as specified above using sulfuric acid (95 to 96 percent) at a temperature of 25° to 30° C. Natural silk fibers dissolve quickly in sulfuric acid while animal fibers require that the acid be heated or treated longer before they will go into solution. If not more than a few fine hair-like fibers remain undissolved in each of the above-mentioned tests, report foreign materials as "none".

4.3.2 Ether extract. Remove approximately 20 grams of the cord, taking not less than 2 grams from each piece comprising the sample, and weigh accurately. Place the weighed sample of cord in a ground-glass Soxhlet extraction apparatus. Pour ether through the condenser until it siphons over into a tared receiving flask. Heat the flask by means of a steam bath and allow the extraction to proceed for 2 hours. Take the apparatus apart at such a time that the ether shall nearly all have collected in the middle section of the apparatus, from which it is caused to siphon over into a receptacle for waste ether. Remove the excess ether in the flask by means of a current of carbon dioxide free air. Dry the flask at 100° C for 1 hour, cool in a desiccator, and weigh. Calculate the grain in weight as percent ether extract.

4.3.3 Acidity.

4.3.3.1 pH of water extract. Weigh accurately a portion of approximately 5 grams of the sample, taking at least 0.5 gram from each piece comprising the sample, and place in a beaker with 150 ml of distilled water. Boil the water from 30 to 45 minutes. Bring the solution back to its original volume by the addition of freshly boiled distilled water, allow to cool, and determine the pH of the water extract by means of an electrometer or pH meter using a glass electrode. Retain the solution for the determination of organic acidity or alkalinity.

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4.3.3.2 Organic acidity or alkalinity. Filter the solution obtained as specified in 4.3.3.1. If the pH of the aqueous solution is 7.0 or greater, titrate with N/10 HCl using methyl red as the indicator. If the pH is less than 7.0 titrate with N/10 NaOH using phenolphthalein as the indicator. Calculate any acidity to percent acetic acid as follows:

$$\text{Percent acetic acid} = \frac{6.0\ VN}{W}$$

Calculate any alkalinity to percent sodium carbonate as follows:

$$\text{Percent sodium carbonate} = \frac{5.3\ VN}{W}$$

where

V = milliliters of standard acid or alkali used in titration
N = normality of standard acid or alkali used
W = grams of sample.

4.3.4 Ash. Tare a porcelain crucible and transfer to this an accurately weighed portion of about 3 grams of the sample. Place the crucible in a muffle furnace or heat over a blast burner for 2 hours. Cool and weigh. Calculate the gain in weight as percent ash.

4.3.5 Breaking strength. Cut one or more test specimens 36 inches in length from each piece of the sample. Test those specimens by means of a standard textile tensile strength machine of an inclination balance or pendulum type of such capacity that the arm makes an angle between 4° and 45° with the vertical when rupture occurs, and having a rate of separation of the grips of 12 ± 2 inches per minute under no load. Use smooth grips of the drum type, 2 inches in diameter. At the beginning of the test the distance between the grips shall be not less than 6 inches. In case a specimen breaks less than one-half inch from the point at which it touches the drum grips or at any point touching the grips, disregard the result and test another specimen from the same piece. Calculate the average of the results.

4.3.6 Yards per pound. Measure 10 yards from each piece in the composite sample and weigh these together on a balance sensitive to 0.1 gram. Calculate the number of yards per pound.

$$\text{Yards per pound} = \frac{453.6A}{B}$$

where

A = number of yards weighed
B = weight of pieces, in grams.

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4.3.7 Stretch. Place two marks, 12 inches apart, on each of three test specimens while stretched with a load of 1 pound. Subject the specimens to a force of 50 pounds for 10 minutes and measure the distance between the two marks. Calculate the increase in distance between the two marks as percent stretch. The average percent stretch of the three specimens shall conform to that specified in 3.3.1.

4.4 Resubmission. If the lot fails to pass the inspection tests, the manufacturer shall have the option of having additional tests made without expense to the Government to effect complete segregation of all defective portions of the lot. The manufacturer may then remove or replace defective portions of the lot and resubmit the lot for acceptance provided that the markings on the containers are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the procuring activity.

5. PACKAGING

5.1 Preservation-packaging. Each cone of silk lacing cord shall be wrapped separately with a neutral grade of wrapping paper which will protect the silk from moisture and sunlight.

5.2 Packing. Unless otherwise specified by the procuring activity, the wrapped cones of silk lacing cord shall be packed in the form of bales wrapped in burlap or shall be packed in wooden boxes. No bale or box shall contain more than 200 pounds of cord. Containers shall conform to Consolidated Freight Classification Rules for lowest rates.

5.3 Marking. In addition to any special marking required by the contract (see 6.1), shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Type and class of cord required (see 1.2)
c. Size of lot (see 4.2.1)
d. Special marking, if required (see 5.3).

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6.2 Because of the extensiveness of changes, asterisks are not used in this revision to identify changes with respect to the previous revision.

Preparing activity:

Navy - OS
(Project No. 4020-N013)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-453C(OS) Cord, Lacing, Silk (FOR ASSEMBLY OF PROPELLING CHARGES)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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